Math Inquirers Primary 6 (after the PSLE): Maths and Coding Bridge: Scheme of Work

Holiday program for ages 11 to 13, after the PSLE • 10 lessons of 90 minutes • 2 lessons a week for 5 weeks • live online, up to 8 children

Approach. Enrichment, not extra drill: non-routine puzzles on topics chosen to complement the school syllabus. Lessons 1 to 9 follow the same routine, in minutes of the 90: Opening (2), sneak-peek puzzles (3.5 to 4.5), warm-up (5.5 to 6.5), explore (12.5 to 15), mini-lesson (8 to 12.5), brain break (4.5 to 5.5), worksheet (30 to 35.5), flex (1.5 to 4), talk time (5 to 6), exit slip (2.5 to 4.5), looking back (2), and close and stamp (2). Lesson 10 has its own plan. The time plan under each lesson is the one in the tutor guide.
Habits. Notice it. Test it. Explain why.

Lesson	Title and school topic	Objective	Skills and heuristics practised	Exit ticket
1 Week 1A	**Code the Pattern** Sequences and the nth Term	Continue a sequence, find the rule for the nth term of a sequence with a constant difference, use it to find far terms such as the 100th, work backwards to decide whether a number is in a sequence, and test predictions with a loop and a variable in Scratch.	Pattern spotting, generalising with a letter, predicting, testing in code, debugging (counting the jumps)	The sequence 8, 11, 14, 17, ... (a) Write the rule for the nth term. (b) Find the 100th term. (c) A loop starts at the 1st term. How many times must it repeat "change by 3" to reach the 100th term?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4** · Warm-up **6.5** · Explore **12.5** · Mini-lesson **11** · Brain break **4.5** · Worksheet **33.5** · Flex **2.5** · Talk time **5** · Exit slip **4.5** · Looking back **2** · Close and stamp **2**			
2 Week 1B	**Prime Hunters** Primes, Factors and Prime Factorisation	Find all the factors of a number using factor pairs, decide whether a number is prime by testing small divisors, write a number as a product of primes with a factor tree, use mod (the remainder), if and a counter in code to count factors, and explore Goldbach pairs and numbers with exactly 3 factors.	Systematic listing (factor pairs), testing and ruling out, reasoning about when to stop, tracing code, generalising (squares of primes)	(a) Is 111 prime? Show your test. (b) Write 66 as a product of primes. (c) Optional: is 133 prime?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **6** · Explore **14** · Mini-lesson **8.5** · Brain break **5** · Worksheet **35.5** · Flex **1.5** · Talk time **5** · Exit slip **4** · Looking back **2** · Close and stamp **2**			
3 Week 2A	**Below Zero** Negative Numbers	Order and compare negative numbers on a number line; add and subtract with negative numbers by moving along the line, in temperature, lift and sea-level contexts; use "change by" a negative number in code and trace a variable that goes below 0.	Number line reasoning, splitting a jump at 0, difference as a distance, tracing code, debugging, planning a loop	It is −3 °C. The temperature falls 6 degrees, then rises 4 degrees. What is the temperature now?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **3.5** · Warm-up **6** · Explore **15** · Mini-lesson **10** · Brain break **5** · Worksheet **33** · Flex **3** · Talk time **6** · Exit slip **2.5** · Looking back **2** · Close and stamp **2**			
4 Week 2B	**Algebra Machines** Expressions and Equations	Find the rule of a function machine from a table and write it with a letter (OUT = 3n + 2); see that the order of the steps matters and use brackets; solve simple equations by running the machine backwards and checking forwards; build an input-rule-output program and use a search loop to test an answer.	Rule finding from a table, writing expressions with a letter, order of operations and brackets, inverse operations, solving simple equations, tracing loops, explaining a number trick with algebra	A machine does × 4, then + 5. (a) IN 6: what is OUT? (b) OUT 45: what was IN?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4** · Warm-up **6.5** · Explore **14** · Mini-lesson **8** · Brain break **5.5** · Worksheet **34.5** · Flex **3.5** · Talk time **5** · Exit slip **3** · Looking back **2** · Close and stamp **2**			
5 Week 3A	**Every Possible Way** Counting	Count every possible case without missing one or counting one twice: list in order, use the multiplication rule for choices made one after another, take care with order and with 0 in digit puzzles, count handshakes and grid routes, and test each count with nested loops in code.	Systematic listing, the multiplication rule, avoiding double counting, adding as you go (grid routes), tracing and planning nested loops	Six friends meet. Every pair of friends shakes hands once. How many handshakes are there? Show how you know.
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **6.5** · Explore **13** · Mini-lesson **9.5** · Brain break **5** · Worksheet **34.5** · Flex **3** · Talk time **5** · Exit slip **3** · Looking back **2** · Close and stamp **2**			

Lesson	Title and school topic	Objective	Skills and heuristics practised	Exit ticket
6 Week 3B	**Fair or Unfair?** Probability	Write a probability as a fraction (ways it can happen out of equally likely outcomes), list all 36 outcomes for two dice, decide whether a game is fair, change a game to make it fair, and run a game 1000 times in code, explaining why the result is close to what we expect but not exact.	Listing equally likely outcomes, fractions in simplest form, expected number = fraction × number of tries, judging fairness, simulating with random numbers	Roll two dice and add. What is the probability that the total is 8? If you roll them 360 times, about how many 8s do you expect?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **6** · Explore **14.5** · Mini-lesson **10.5** · Brain break **5** · Worksheet **31.5** · Flex **2.5** · Talk time **6** · Exit slip **3.5** · Looking back **2** · Close and stamp **2**			
7 Week 4A	**Turtle Geometry** Angles in Polygons	Use the idea that a turtle going once round a closed shape turns 360° altogether to find the turn angle (360° ÷ sides) and inside angle (180° – turn) of regular polygons, work backwards from an angle to the number of sides, extend to star polygons (turn = 360° × times round ÷ points), and test every prediction with a pen program in Scratch (or Python turtle).	Turn (exterior) and inside (interior) angles, angles on a straight line, angle sum of a polygon, star polygons, predicting and tracing code, repeat loops and variables	A turtle draws a regular polygon and turns 5° at every corner. How many sides does the polygon have? What is each inside angle?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **6.5** · Explore **14.5** · Mini-lesson **12.5** · Brain break **5.5** · Worksheet **30** · Flex **2.5** · Talk time **5** · Exit slip **3** · Looking back **2** · Close and stamp **2**			
8 Week 4B	**Secret Codes** Clock (Modular) Arithmetic	Do clock arithmetic (counting past the end wraps round to the start), use n mod m as the remainder when n is divided by m, solve clock and calendar puzzles with mod 12 and mod 7, code and decode Caesar ciphers with letter numbers (A = 0 … Z = 25, code = (letter + shift) mod 26), crack a code, and test it all with a Scratch program that uses a list of letters and mod.	Remainders and mod, clock and calendar arithmetic, negative numbers on a clock, Caesar ciphers, cracking codes by reasoning, lists and letters in code, tracing code in a table	Code the word HI with a shift of 25. What do you notice?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4** · Warm-up **5.5** · Explore **14.5** · Mini-lesson **11** · Brain break **5** · Worksheet **31** · Flex **4** · Talk time **6** · Exit slip **3** · Looking back **2** · Close and stamp **2**			
9 Week 5A	**Data Detectives** Averages and Data	Find the mean as total ÷ number of values and as levelling out, use total = mean × number of values to find a missing value, predict how changing the data changes the mean, compare the mean with the median and mode when one value is extreme, and build a list program that adds up the data and finds the mean.	Mean, working back from a mean to a total, reasoning about change, median and mode, lists and loops in code, tracing code	The mean of 3 numbers is 9. Two of them are 5 and 12. What is the third number? A fourth number, 13, is added. What is the new mean?
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4.5** · Warm-up **5.5** · Explore **14** · Mini-lesson **11.5** · Brain break **5.5** · Worksheet **32.5** · Flex **1.5** · Talk time **5** · Exit slip **4** · Looking back **2** · Close and stamp **2**			
10 Week 5B	**Showcase Challenge** Mixed: Maths and Code	Use ideas from all nine lessons (sequences, primes, negative numbers, algebra, counting, probability, angles, clock arithmetic, averages) in a mixed challenge round, then plan, build, test and present a small puzzle program, working out the maths by hand first and explaining why it works.	Choosing a strategy, connecting topics, planning code, testing with a known answer, presenting and explaining	Write one thing your program does, one test you did (the input and the answer you expected), and the maths that explains it.
	Time plan (minutes): Opening **2** · Sneak-peek puzzles **4** · Challenge round (teams of 2) and check **13.5** · Starter projects (A-D) **7** · Brain break **3.5** · Plan and build **28** · Flex **2.5** · Showcase (8 children) **22** · Exit slip and talk question **3.5** · Looking back **2** · Certificates and close **2**			

Assessment and progress

Each lesson has a 2-page Start here worksheet, then Think harder, Go further and an Investigation with a Look-back (printed for every child, who may choose any of them at any time; nobody is placed in a group), an exit ticket and an Inquiry Journal stamp. Tutors record S (secure), D (developing) or H (needs help) for each child on the progress tracker in the tutor guide. Parents receive a short note each week and the Weekly Dose of Thinking Math practice pack.

Materials

Printed student workbook with cut-outs, pencil, scissors and paper; a laptop or tablet with a camera. Tutors use the slides for each lesson and the tutor guide scripts.